Kevin Abraham Wilson

GPU Computing & ML Systems

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Education

Mar Baselios College of Engineering and Technology (Autonomous) Thiruvananthapuram, Kerala
B.Tech — Computer Science and Engineering (Artificial Intelligence) *Expected June 2027*
CGPA: 8.20/10
Class Representative *Dec 2023 – Feb 2026*
Relevant Coursework: Deep Learning, Numerical Linear Algebra, Computer Architecture, Algorithms Analysis & Design, Operating Systems

Academic & Research Interests

B.Tech Computer Science student focused on GPU computing, deep learning systems, and hardware-accelerated inference. Developing CUDA-based inference systems and custom GPU kernels, with hands-on experience in PyTorch, TensorRT, profiling, and performance analysis.

Technical Skills

Programming: Python, C++, CUDA C++, Bash
Deep Learning: PyTorch, PyTorch Lightning, TensorFlow
GPU & Inference: CUDA, TensorRT, ONNX, Nsight Compute, Nsight Systems
Deployment & Systems: Linux (Ubuntu), Docker, FastAPI
Developer Tools: Git, GitHub, Weights & Biases, CMake

Selected Projects

Llama Transformer Runtime | Final Year Project *May 9, 2026 – Sept 15, 2026*
CUDA (SM 89+) • C++ • PyTorch • Nsight Compute • Nsight Systems • Linux

- Built a bare-metal C++/CUDA inference runtime from scratch to execute the **Llama 3.2 1B Instruct** model (BF16) on Ampere/Hopper architectures, achieving a footprint of only **2,573 MB** runtime memory.
- Wrote hand-optimized computational kernels—including **FlashAttention v2** achieving **34.924 TFLOP/s** at 32.44% occupancy—utilizing **WMMA/PTX hardware intrinsics** and **cp.async** double-buffering.
- Sustained an end-to-end inference throughput of **606.403 tokens/s** with a peak tokenizer speed of **2.378 M tokens/s**; verified mathematical correctness with a strict **0.99995 cosine similarity** against the PyTorch reference.

Custom FlashAttention-2 CUDA Forward Kernel *Aug 9, 2026 – Sept 15, 2026*
CUDA C++ • GPU Memory Hierarchy • Shared Memory Tiling • Online Softmax

- Developed a hardware-optimized **FlashAttention-v2 forward kernel** from scratch in CUDA C++, using tiled attention to completely eliminate $N \times N$ global memory traffic.
- Optimized execution targeting NVIDIA Ada Lovelace (SM 8.9), implementing **4-warp split-Q parallelism**, Tensor Core `mma.sync`, register-resident values, and `cp.async` double-buffering.
- Achieved **34.924 TFLOP/s** at sequence length 2048 on an RTX 4060 Laptop GPU for BF16 causal attention with **GQA** and online softmax; validated memory safety via **Compute Sanitizer**.

GPU-Accelerated CNN Inference Pipeline *Jan 3, 2026 – Jan 17, 2026*
CUDA • TensorRT • ONNX • PyTorch • Python

- Built an end-to-end EfficientNet-B0 inference pipeline on **RTX 4060**, achieving **83.4% CIFAR-100 Top-1 accuracy** and **28% higher throughput (1,076 → 1,387 images/sec)** with TensorRT.
- Optimized PyTorch → ONNX → TensorRT inference, reducing batch-128 latency by **22.4% (118.93 → 92.27 ms)** with no accuracy loss.
- Benchmarked inference across **batch sizes 1–128**, characterizing latency/throughput scaling and identifying hardware saturation points for GPU deployment.